

NANDHAKUMAR M

Full Stack AI Engineer

Coimbatore, Tamil Nadu, India · Open to relocation to Canada

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PROFESSIONAL SUMMARY

Full stack software engineer with 5+ years of experience building enterprise web applications, data platforms and Generative AI solutions for healthcare, business continuity and analytics. Skilled in Python (FastAPI, Django), React, Node.js, PostgreSQL, Azure and AWS. Recent work includes LLM-powered applications with OpenAI and Ollama, AI agents that call tools through the Model Context Protocol, and document intelligence with OCR. Recognized as Best Performer at Firstsource Solutions.

TECHNICAL SKILLS

AI & GenAI: OpenAI API, Azure OpenAI, Ollama, LLM tool calling, Model Context Protocol (MCP), prompt engineering, sentiment analysis, OCR (Tesseract)

Backend: Python (FastAPI, Django REST Framework), Node.js, Express, Java (Spring Boot), REST APIs, microservices

Frontend: React, Next.js, TypeScript, JavaScript, React Native (Expo), Redux Toolkit, Tailwind CSS

Data: PostgreSQL, MySQL, MongoDB, Redis, PySpark, Microsoft Fabric Lakehouse, Azure Data Lake Storage

Cloud & DevOps: Azure, AWS, Docker, CI/CD, Azure DevOps, Git

Practices: JWT, OAuth 2.0 and SSO, automated testing (pytest, Vitest, Playwright), Agile delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Apr 2025 – Present

Firstsource Solutions Limited, Hyderabad, India

â€¢ Developed Python (PySpark) data pipelines on Microsoft Fabric Lakehouse for large-scale structured and semi-structured data.

â€¢ Reduced ETL processing time by **40%** and increased data reliability by streamlining data workflows.

â€¢ Built interactive dashboards that improved data accessibility and business insight by **35%**.

â€¢ Integrated Azure Data Lake Storage for ingestion, processing and archival of business-critical data.

â€¢ Contributed features to React and Node.js applications and their backend APIs.

Junior Software Engineer

Oct 2021 – Mar 2025

Firstsource Solutions Limited, Hyderabad, India

â€¢ Led full-cycle development of Information Lifecycle Management, Business Continuity Management and healthcare (NGO) platforms with **100%** on-time delivery.

â€¢ Developed Spring Boot microservices and React applications, improving system efficiency by **35%**.

â€¢ Reduced infrastructure costs by **25%** by optimizing AWS and Azure resource usage.

â€¢ Improved query performance by **30%** by redesigning PostgreSQL schemas.

â€¢ Automated Docker deployments to AWS (**40%** faster) and streamlined CI/CD for a **50%** faster release cycle.

â€¢ Wrote API and architecture documentation that reduced new-developer onboarding time by **40%**.

Full Stack Developer

Mar 2021 – Sep 2021

Invious Technologies, Coimbatore, India

â€¢ Led a cross-functional team and increased project delivery speed by **35%** through clear planning.

â€¢ Optimized the Hyper App project, improving system performance by **30%**.

â€¢ Built Spring Boot REST APIs that reduced API response time by **25%**.

â€¢ Managed AWS deployments and cut infrastructure costs by **15%**.

â€¢ Integrated MySQL and MongoDB, improving data retrieval speed by **40%**.

Career break COVID-19 pandemic	Jun 2020 – Feb 2021
Java Web Developer Smartway Industrial Automation, Coimbatore, India	Dec 2019 – May 2020
â€¢ Developed a real-time monitoring dashboard with Java and JSP, improving user accessibility by 20% . â€¢ Improved application performance by 20% through code-level optimization.	

SELECTED PROJECTS github.com/Nandhu29696

AI Email Assistant Python, FastAPI, PostgreSQL, Redis, Next.js, OpenAI
Classifies, prioritizes and summarizes Gmail and Outlook email with an LLM and sentiment and emotion analysis; drafts replies, routes messages to teams and sends real-time alerts over WebSockets.
JobAgent AI Python, FastAPI, React, Ollama
Career assistant that scores resume-to-job match, generates tailored cover letters and tracks applications.
MCP Healthcare Appointment Agent Node.js, Express, PostgreSQL, Ollama
AI agent that selects and calls typed tools (patient search, doctor availability, booking) to book appointments from natural-language requests.
Loan Document Extraction Django REST Framework, PyMuPDF, Tesseract OCR, React
Extracts structured fields from scanned mortgage documents into a searchable loan tracker.
HealthCamp Management Platform Node.js, Express, PostgreSQL, Azure Blob Storage
Platform for an NGO’s community health camps: camp planning, patient records, bulk data import and reports. In production at hms.nirmaan.org.

CERTIFICATIONS & TRAINING

GitHub Copilot Certification (Exam GH-300) , Microsoft	Valid 2026 – 2028
Develop AI Apps and Agents on Azure (AI-103T00-A course) , Microsoft	Valid 2026 – 2027

EDUCATION

Master of Computer Applications (MCA)	2020
Hindusthan College of Engineering and Technology (Anna University), Coimbatore, India	
Bachelor of Computer Applications (BCA)	2016
Hindusthan College of Arts and Science, Coimbatore, India	

AWARDS

Best Performer , Firstsource Solutions: for technical excellence and contributions to project delivery
Agility and Humanness Award , Firstsource Solutions: for adaptability and collaboration on high-impact projects